

We use the projection map $\pi : \mathbb{R}^{2d} \longrightarrow \mathbb{R}^d$ given by $\pi\left(\begin{smallmatrix} \mathbf{x} \\ \mathbf{x}' \end{smallmatrix}\right) := \mathbf{x} + \mathbf{x}'$, whose dual map $\pi^* : (\mathbb{R}^d)^* \longrightarrow (\mathbb{R}^{2d})^*$ is the diagonal map $\pi^*(\mathbf{c}) = (\mathbf{c}, \mathbf{c})$, by

$$\pi^*(\mathbf{c})\left(\begin{smallmatrix} \mathbf{x} \\ \mathbf{x}' \end{smallmatrix}\right) = \mathbf{c}\left(\pi\left(\begin{smallmatrix} \mathbf{x} \\ \mathbf{x}' \end{smallmatrix}\right)\right) = \mathbf{c}(\mathbf{x} + \mathbf{x}') = (\mathbf{c}, \mathbf{c})\left(\begin{smallmatrix} \mathbf{x} \\ \mathbf{x}' \end{smallmatrix}\right).$$

Now we can write the Minkowski sum as a projection of the product:

$$P + P' := \pi(P \times P').$$

Thus, putting together very simple observations, we get the normal fan of a Minkowski sum. (Careful: it is *not* the direct sum of the fans of the factors!)

Proposition 7.12. *The normal fan of a Minkowski sum is the common refinement of the individual fans:*

$$\mathcal{N}(P + P') = \mathcal{N}(P) \wedge \mathcal{N}(P').$$

Proof.

$$\begin{aligned} \mathcal{N}(P + P') &= \mathcal{N}(\pi(P \times P')) \stackrel{\pi^*}{\cong} \mathcal{N}(P \times P') \Big|_{\pi^*(\mathbb{R}^d)^*} = \\ &= (\mathcal{N}(P) \oplus \mathcal{N}(P')) \Big|_{\pi^*(\mathbb{R}^d)^*} \cong \mathcal{N}(P) \wedge \mathcal{N}(P'), \end{aligned}$$

using Lemma 7.7 for the direct sums, Lemma 7.11 for the projection, and the dual map $\pi^*(\mathbf{c}) = (\mathbf{c}, \mathbf{c})$. $\square$

7.3 Zonotopes

Zonotopes are special polytopes that can be viewed in various ways: for example, as projections of cubes, as Minkowski sums of line segments, and as sets of bounded linear combinations of vector configurations. Each of these descriptions gives different insight into the combinatorics of zonotopes. The following includes several such descriptions, all of which lead us to the same “associated” system of sign vectors that describes the combinatorics of a zonotope. The main goal will be to see in what sense zonotopes and arrangements can be considered equivalent, and how the combinatorial structure of a zonotope is given by an oriented matroid.

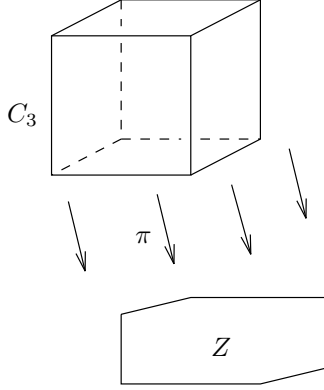
After the general discussion of projections in the last section we now consider a very special (but interesting) case: projections of cubes, that is, $\pi : P \longrightarrow Q$, $\mathbf{x} \longmapsto V\mathbf{x} + \mathbf{z}$ is an arbitrary (surjective) affine map, but P is the d -cube,

$$P = C_p = \{\mathbf{x} \in \mathbb{R}^p : -1 \leq x_i \leq 1 \text{ for all } i\}.$$

Definition 7.13. A *zonotope* is the image of a cube under an affine projection, that is, a d -polytope $Z \subseteq \mathbb{R}^d$ of the form

$$\begin{aligned} Z = Z(V) &:= V \cdot C_p + \mathbf{z} = \{V\mathbf{y} + \mathbf{z} : \mathbf{y} \in C_p\} \\ &= \{\mathbf{x} \in \mathbb{R}^d : \mathbf{x} = \mathbf{z} + \sum_{i=1}^p x_i \mathbf{v}_i, -1 \leq x_i \leq 1\} \end{aligned}$$

for some matrix (vector configuration) $V = (\mathbf{v}_1, \dots, \mathbf{v}_p) \in \mathbb{R}^{d \times p}$.



Equivalently, since every d -cube C_d is a product of line segments $C_d = C_1 \times \dots \times C_1$, we get that every zonotope is the Minkowski sum of a set of line segments. In fact, if π is linear we get

$$\begin{aligned} Z(V) &= \pi(C_1 \times \dots \times C_1) \\ &= \pi(C_1) + \dots + \pi(C_1) \\ &= [-\mathbf{v}_1, \mathbf{v}_1] + \dots + [-\mathbf{v}_p, \mathbf{v}_p], \end{aligned}$$

and thus $Z(V) = [-\mathbf{v}_1, \mathbf{v}_1] + \dots + [-\mathbf{v}_p, \mathbf{v}_p] + \mathbf{z}$ for an affine map given by $\pi(\mathbf{y}) = V\mathbf{y} + \mathbf{z}$.

In the following we will usually assume that $Z = -Z$ is centrally symmetric with respect to the origin $\mathbf{0}$, corresponding to a linear map $\pi : C_n \longrightarrow Z$.

Example 7.14. By definition, the cubes C_d are zonotopes, where the projection map can be taken to be the identity.

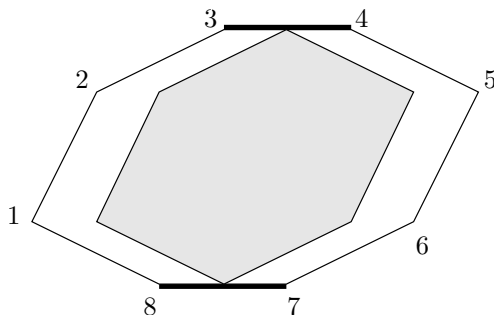
Also, every centrally symmetric, 2-dimensional $2p$ -gon $P_2(2p)$ arises as the projection of a p -cube to the plane. In fact, if the vertices of $P_2(2p)$ are

$$\mathbf{x}_1, \dots, \mathbf{x}_p, \mathbf{x}_{p+1}, \dots, \mathbf{x}_{2p}$$

in cyclic order, with $\mathbf{x}_{p+i} = -\mathbf{x}_i$, then we get

$$P_2(2p) = \left[-\frac{\mathbf{x}_2 - \mathbf{x}_1}{2}, \frac{\mathbf{x}_2 - \mathbf{x}_1}{2}\right] + \dots + \left[-\frac{\mathbf{x}_{p+1} - \mathbf{x}_p}{2}, \frac{\mathbf{x}_{p+1} - \mathbf{x}_p}{2}\right].$$

One way to prove this is by induction on p , by taking any pair of opposite (parallel, of same length) edges, and showing that it corresponds to a Minkowski summand of $P_2(2p)$.



We invite the reader to provide his or her own proof.

Example 7.15. The permutahedron Π_{n-1} (Example 0.10) is a zonotope of dimension $d = n - 1$, arising from an affine projection of the cube of dimension $p = \binom{n}{2}$:

$$\begin{aligned} \Pi_{n-1} = & \frac{n+1}{2} \mathbf{1} + \left[-\frac{e_2 - e_1}{2}, \frac{e_2 - e_1}{2} \right] + \left[-\frac{e_3 - e_1}{2}, \frac{e_3 - e_1}{2} \right] + \dots \\ & \dots + \left[-\frac{e_n - e_{n-1}}{2}, \frac{e_n - e_{n-1}}{2} \right]. \end{aligned}$$

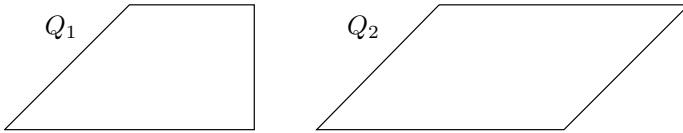
Perhaps the easiest way to see that this Minkowski sum yields the right polyhedron is first to observe that it is invariant under permutation of coordinates, and then compute the points of the sum that maximize a linear function $\mathbf{c} \in (\mathbb{R}^n)^*$ with $c_1 < c_2 < \dots < c_n$: this is easily seen to be the vertex

$$\begin{aligned} \mathbf{v} &= \frac{n+1}{2} \mathbf{1} + \frac{e_2 - e_1}{2} + \frac{e_3 - e_1}{2} + \dots + \frac{e_n - e_{n-1}}{2} \\ &= \frac{n+1}{2} \mathbf{1} - \frac{n-1}{2} e_1 - \frac{n-3}{2} e_2 - \dots - \frac{n+1-2n}{2} e_n = \begin{pmatrix} 1 \\ 2 \\ \vdots \\ n \end{pmatrix}. \end{aligned}$$

There are a few more “obvious” properties of zonotopes: for example, all zonotopes are centrally symmetric. Also, since every face of a cube is a (translated) cube, we get that every face of a zonotope is again a zonotope, and thus centrally symmetric with respect to its barycenter. This property characterizes zonotopes. In fact, any polytope all whose 2-faces are centrally symmetric is a zonotope; see Bolker [120] and the references given there, Schneider [475], or [96, Prop. 2.2.14]. Even stronger: any polytope whose k -faces are all centrally symmetric, for some k with $2 \leq k \leq d - 2$,

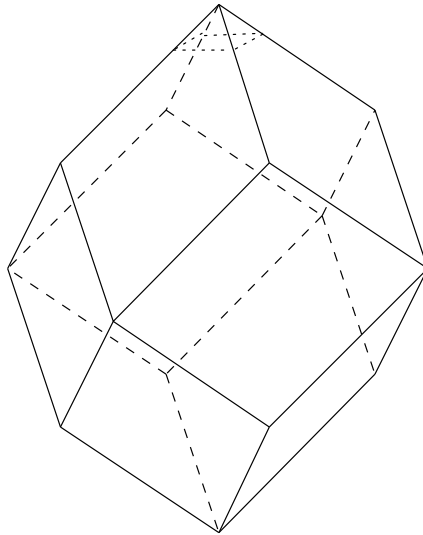
is a zonotope, by McMullen [388]. The same fails if only all the facets ($k = d-1$) are centrally symmetric: in this case we have a counterexample in the regular 24-cell (a sporadic 4-dimensional regular polytope, described for example in Coxeter [164, Sect. 8.2]), whose facets are all regular octahedra; counterexamples for all $d \geq 4$ can be found in McMullen's paper [393].

In particular, we see that being a zonotope is a geometric property, not a combinatorial one. For example, the quadrilateral Q_1 is not a zonotope, but the quadrilateral Q_2 is.



Thus being a zonotope is preserved under affine equivalence (in fact, under affine projections), but not in general under combinatorial equivalence.

It may require a second of thought to figure out that in general zonotopes are not simple polytopes (though the permutahedra are). Our next picture shows a zonotope generated by four line segments in $\mathbb{R}^3$, no three of them coplanar. The resulting zonotope ($d = 3$, $p = 4$) has 8 simple vertices of degree 3, and 6 nonsimple vertices of degree 4. The figure indicates (by a dotted line) that the vertex figure of the top vertex is a square.



For the combinatorial structure of zonotopes, we have Lemma 7.10: the faces of Z can be uniquely associated with the faces of the cube it is projected from.

Now every nonempty face F of the p -cube can be associated with a sign vector. Here the natural construction associates with F a row vector

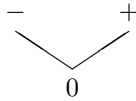
$\sigma(F) \in (\{+, -, 0\}^p)^*$, for example by

$$\sigma(F) = \text{sign}(\text{int}(F^\diamond)).$$

This is well defined, since the polar of a nonempty face of the cube is a proper face of the crosspolytope, which has constant signs on the interior. There are other, equivalent, ways to relate a face F with the sign vector $\sigma = \sigma(F) \in (\{+, -, 0\}^p)^* \equiv (\{+1, -1, 0\}^p)^*$, for example by

$$\begin{aligned} F &= \left\{ \sum_{i=1}^p \lambda_i \mathbf{e}_i : \begin{array}{l} \lambda_i = +1 \text{ for } \sigma_i = +, \\ \lambda_i = -1 \text{ for } \sigma_i = -, \\ -1 \leq \lambda_i \leq +1 \text{ for } \sigma_i = 0 \end{array} \right\} \\ &= \{ \mathbf{x} \in C_p : x_i = \sigma(F)_i \text{ for all } i \text{ with } \sigma(F)_i \neq 0 \}. \end{aligned}$$

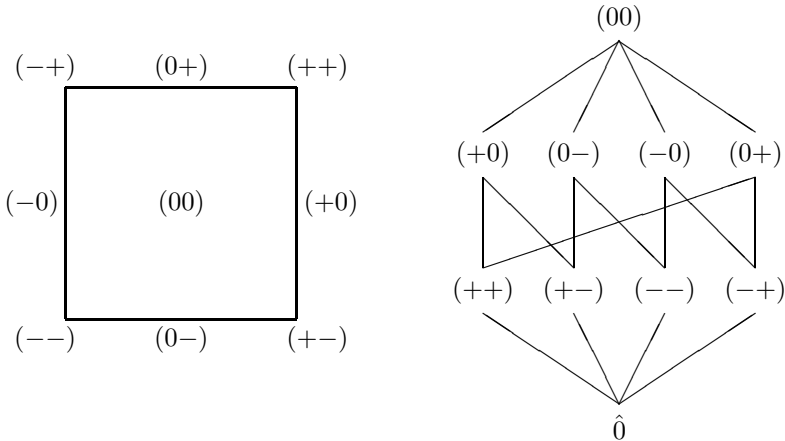
Recall from Section 6.3(a) the componentwise partial order on sign vectors, which is induced by



We find that the smaller a face F of C_p , the larger its sign vector $\sigma(F)$ will be in this partial order “ $\leq$.” Also, in order to obtain the whole face lattice we have to add an extra minimal element, since the empty face does not contain points, and it does not have an associated sign vector, either. Thus, we get

$$(L(C_p), \subseteq) \cong \{\hat{0}\} \cup ((\{+, -, 0\}^p)^*, \geq).$$

The following sketch shows the signs associated with the faces of C_2 , and the face lattice $L(C_2)$ with the signs corresponding to its elements.



With this we get sign vectors not only for the faces of cubes, but also for the faces of zonotopes: if $\pi : C_p \rightarrow Z$ is the projection that defines the zonotope Z , then for every nonempty face $G \in L(Z)$ of the zonotope we get the nonempty face $\pi^{-1}(G) \in L(C_d)$, and thus we put

$$\sigma(G) := \sigma(\pi^{-1}(G)) = \text{sign}(\mathbf{x}) \in (\{+, -, 0\}^p)^*,$$

where $\mathbf{x}$ is an arbitrary point of $\pi^{-1}(G)$ — for example, the center.

From this we get a sign for every face of the zonotope, and we have $\sigma(G) \leq \sigma(G')$ if and only if $G \supseteq G'$: thus the face lattice of the zonotope is entirely determined by the system of sign vectors, and antiisomorphic to it as a poset:

$$(L(Z), \subseteq) \cong \hat{0} \cup \left(\{ \sigma(G) : G \in L(Z) \setminus \{\emptyset\} \}, \geq \right),$$

where $\hat{0}$ is defined to be smaller than any sign vector $\sigma(G)$. Caution: *the partial order on the face lattices is opposite to the order on sign vectors. The larger a face of Z , the smaller a sign vector we associate with it*, in the partial order on sign vectors induced by $0 < +$ and $0 < -$.

This assignment of sign vectors to the faces of a zonotope may look a little mysterious. In particular, the description that we have given (from the projection of a cube) does not tell us much about the structure of the collections of sign vectors that we get, and how they are related to the matrix V that defines the projection. Thus, we take a “fresh start” here, and obtain the same sign vectors from a different approach, via optimization.

The following theorem is so basic that we give two proofs. The first proof shows how a sign vector $\sigma_C \in \{+, -, 0\}^p$ is associated with every cone in the normal fan $\mathcal{N}(Z)$ of the zonotope. Similarly, the second proof obtains a sign vector $\sigma(G) \in \{+, -, 0\}^p$ for every face $G \in L(Z)$ — the same sign vector that we have just “pulled down” from the p -cube, of course. Thus after this we have three different constructions of the sign vector system associated with a zonotope.

Theorem 7.16. *Let $Z = Z(V) \subseteq \mathbb{R}^d$ be a zonotope. Then the normal fan of $\mathcal{N}(Z)$ of Z is the fan $\mathcal{F}_A$ of the hyperplane arrangement*

$$\mathcal{A} = \mathcal{A}_V := \{H_1, \dots, H_p\}$$

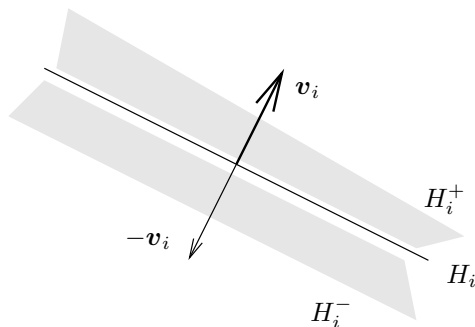
in $\mathbb{R}^d$ given by

$$H_i := \{\mathbf{c} \in (\mathbb{R}^d)^* : \mathbf{c}\mathbf{v}_i = 0\}.$$

First Proof. For each of these hyperplanes we define the *positive halfspace* by

$$H_i^+ = \{\mathbf{c} \in (\mathbb{R}^d)^* : \mathbf{c}\mathbf{v}_i \geq 0\},$$

and the negative one similarly. Thus the normal fan of the single line segment $[-\mathbf{v}_i, \mathbf{v}_i]$, is the set $\{H_i, H_i^+, H_i^-\}$: a hyperplane and the two half-spaces determined by it.



Now we use Proposition 7.12 to see that the normal fan of the Minkowski sum of the line segments $[-\mathbf{v}_i, \mathbf{v}_i]$ is the hyperplane arrangement $\mathcal{A}_V$, that is, the common refinement of the fans of the individual hyperplanes.

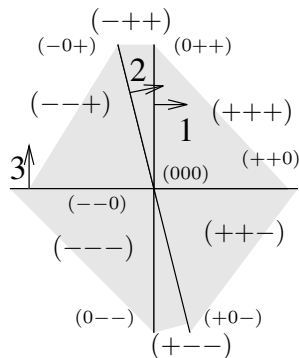
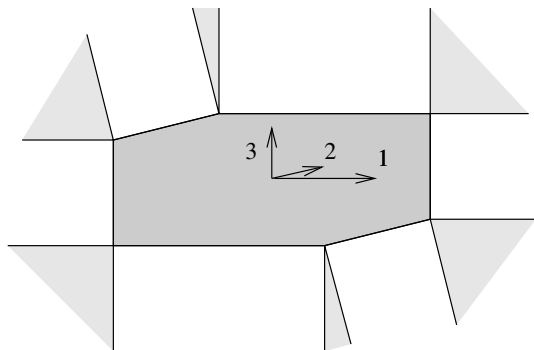
The position of $\mathbf{c}$ with respect to the fan $\{H_i, H_i^+, H_i^-\}$ is determined by the sign of $\mathbf{c}\mathbf{v}_i$. In fact, if $\text{sign}(\mathbf{c}\mathbf{v}_i) = 0$, then $\mathbf{c}$ lies in H_i ; if this sign is $+$, then $\mathbf{c}$ is in the interior of H_i^+ ; and if it is $-$, then $\mathbf{c}$ lies in the interior of H_i^- .

Thus in the common refinement $\mathcal{A}_V$, the position of $\mathbf{c}$ is given by the sign vector

$$\text{sign}(\mathbf{c}V) \in \{+, -, 0\}^p,$$

whose first coordinate records the relative position with respect to H_1 , the second coordinate refers to H_2 , and so on. In particular, we get distinct sign vectors for the distinct cones in $\mathcal{N}(Z)$, and inclusion of cones corresponds to the usual partial order on sign vectors.

Our sketch shows a zonotope, its normal cones (drawn into the same figure with right angles), the normal fan assembled from them (which is a hyperplane arrangement), and the signs that we associate with each of the cones in the normal fan.



Second Proof. Assume that we want to maximize a function $\mathbf{x} \mapsto \mathbf{c}\mathbf{x}$ over the set of all bounded linear combinations

$$Z = \left\{ \sum_{i=1}^n \lambda_i \mathbf{v}_i : -1 \leq \lambda_i \leq +1 \right\}.$$

We can maximize this sum by maximizing each of the summands separately, and thus achieving the maximum on the following face of Z :

$$\begin{aligned} Z^{\mathbf{c}} &= \{ \mathbf{y} \in Z : \mathbf{c}\mathbf{y} = \max_{\mathbf{x} \in Z} \mathbf{c}\mathbf{x} \} \\ &= \left\{ \sum_{i=1}^n \lambda_i \mathbf{v}_i : \begin{array}{ll} \lambda_i = -1 & \text{if } \mathbf{c}\mathbf{v}_i < 0, \\ -1 \leq \lambda_i \leq +1 & \text{if } \mathbf{c}\mathbf{v}_i = 0, \\ \lambda_i = +1 & \text{if } \mathbf{c}\mathbf{v}_i > 0 \end{array} \right\}. \end{aligned}$$

Thus the decision “which face of Z maximizes $\mathbf{c}$ ” is equivalent to the decision, for each i , of whether $\mathbf{c}$ lies on the hyperplane H_i itself, on its negative side, or on its positive side, that is, by the position of $\mathbf{c}$ in the fan of the arrangement $\mathcal{A}$. $\square$

The family of hyperplanes $\mathcal{A}$ thus gives a combinatorial interpretation for the covectors of the configuration $V = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. Here the interesting case is the one where the configuration V spans $\mathbb{R}^d$, such that the zonotope $Z(V)$ has dimension d , and the hyperplane arrangement $\mathcal{A}_V$ is *essential*: the intersection of all the hyperplanes is the origin, $H_1 \cap H_2 \cap \dots \cap H_n = \{\mathbf{0}\}$, and thus the cones in $\mathcal{A}_V$ are all pointed cones.

Corollary 7.17. *Let $V \in \mathbb{R}^{d \times p}$ be a vector configuration in $\mathbb{R}^d$. Then there is a natural bijection between the following three families:*

- (the sign vectors of) the nonempty faces of the zonotope $Z(V) \subseteq \mathbb{R}^d$,
- (the sign vectors of) the faces of the hyperplane arrangement $\mathcal{A}_V$,
- the signed covectors of the configuration V .

Thus we have identifications

$$L(Z(V)^\Delta) \setminus \{\hat{1}\} \longleftrightarrow L(\mathcal{A}_V) \longleftrightarrow \mathcal{V}^*(V) \subseteq (\{+, -, 0\}^p)^*.$$

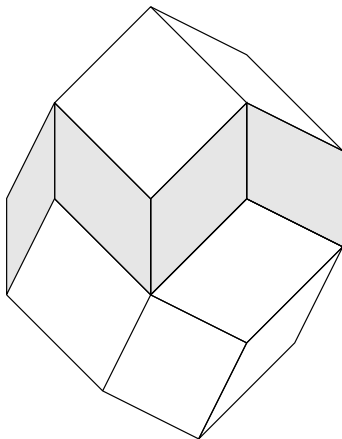
Similarly (assuming that V has full rank), there is a natural bijection between the following three families:

- (the sign vectors of) the facets of the zonotope $Z(V) \subseteq \mathbb{R}^d$,
- (the sign vectors of) the one-dimensional rays of the hyperplane arrangement $\mathcal{A}_V$, and
- the signed cocircuits of the configuration V .

In notation:

$$\text{facets}(Z(V)) \longleftrightarrow \text{vert}(Z(V)^\Delta) \longleftrightarrow \text{rays}(\mathcal{A}_V) \longleftrightarrow \mathcal{C}^*(V).$$

The i th zone of $Z(V)$ is the collection of all faces that have $[-\mathbf{v}_i, \mathbf{v}_i]$ as a Minkowski summand. The zones geometrically form “belts” around the surface of a zonotope, and in fact completely cover it.



These zones may also be held responsible for the name “zonotope.” Under the bijection between faces of a zonotope and cones in its hyperplane arrangement, the i th zone corresponds to the i th hyperplane H_i in the arrangement.

Note that $\mathbf{v}_i$ and $\mathbf{v}_j$ determine the same zone, and the same hyperplane, exactly if they are parallel vectors. This is a degenerate case that one usually excludes from the discussion. In fact, there is an even more degenerate case, if $\mathbf{v}_i = \mathbf{0}$ for some i . In this case $\mathbf{v}_i$ does not contribute to the geometry, we can just ignore it when constructing the zonotope, $H_i = (\mathbb{R}^d)^*$ is the full dual space, and i is a one element circuit (known as a *loop*) in the oriented matroid, which can safely be deleted.

The number of zones is the principal complexity measure for zonotopes: it coincides with the number of different hyperplanes in the associated arrangement, and with the number of equivalence classes of elements in the oriented matroid. If we assume that V is *simple*, that is, there are no zero or parallel vectors in $V \in \mathbb{R}^{d \times p}$, then the number of zones is p . The key observation is that this parameter can be read off directly from the zonotope Z , and does not depend on the choice of V . However, for every zonotope Z there is a simple vector configuration that defines it, and the vector configuration is unique up to permutations and sign changes.

Under the translation from zonotopes to arrangements and back, simple zonotopes correspond to simplicial hyperplane arrangements — a very

classical topic of geometric study because of its relation to the theory of reflection groups and Lie algebras. See the notes at the end of the lecture.

There is also a metrical correspondence, which we get by using polarity. In fact, from the simple observation that the normal fan of a polytope is the face fan of its polar (Exercise 7.1), we get a polytope for every hyperplane arrangement that “spans” the arrangement as a fan.

Corollary 7.18. *Let $\mathcal{A}_V$ be a hyperplane arrangement in $(\mathbb{R}^d)^*$. Then the face fan of the polar of the associated zonotope is given by $\mathcal{A}_V$:*

$$\mathcal{F}_{\mathcal{A}_V} = \mathcal{F}(Z(V)^\Delta).$$

In particular, if V spans $\mathbb{R}^d$, then the arrangement is essential, the zonotope is full-dimensional, and its polar is a polytope.

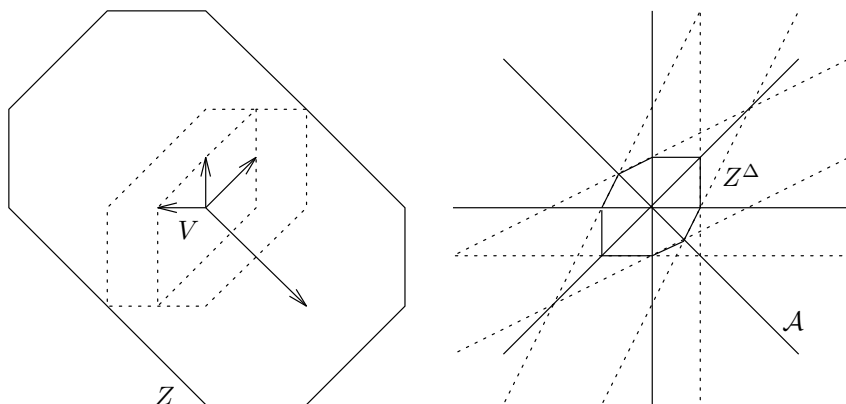
Thus the fan of an essential hyperplane arrangement is always polytopal.

The following constructs an explicit example for this. Note that from our set-up the proof is simple — but the geometric fact that we can construct a polytope that “fits” any given hyperplane arrangement is not obvious at first sight.

Example 7.19. For the matrix

$$V = \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 & -1 \end{pmatrix}$$

we construct



Corollary 7.18 shows that the combinatorics of zonotopes is equivalent to the combinatorics of hyperplane arrangements in a very strong sense. Let us close this section with one nontrivial application of this equivalence. For this we take a classical theorem from the theory of hyperplane arrangements, Shannon’s theorem [493]:

Every essential arrangement $\mathcal{A}$ of n hyperplanes in $\mathbb{R}^d$ has at least $2n$ simplicial regions.

More precisely, adjacent to every hyperplane there are at least $2d$ simplicial regions, and nonadjacent to any given hyperplane there are at least $2(n - d)$ simplicial regions.

(Note that the simplicial regions come in pairs, since the opposite of a simplicial cone is simplicial as well). From Shannon's theorem, we derive the following theorem about zonotopes.

Theorem 7.20 (Shannon's theorem for zonotopes).

Every d -zonotope with n zones has at least $2n$ simple vertices.

More precisely, on every zone there are at least $2d$ simple vertices, and disjoint from any given zone there are at least $2(n - d)$ simple vertices.

The proof for Shannon's theorem for arrangements is not difficult: we refer to Roudneff & Sturmfels [467], where several different proofs are presented, and to [96, Thm. 2.1.5]. The interesting thing is that there is no "entirely combinatorial" proof. In fact, the corresponding statement for oriented matroids is false. As we will see below, this translates into very interesting effects for "zonotopal tilings."

7.4 Nonrealizable Oriented Matroids

We have by now *seen* so many oriented matroids around that you shouldn't be scared any more if you hit their axiomatic definition a few lines down. In fact, the axioms below just describe abstractly the "most important" properties shared by the sign vector systems of — equivalently —

- hyperplane arrangements
- zonotopes
- vector configurations
- affine point configurations (vertices of polytopes!).

We have seen that in each of these cases we get a sign vector system of the form $\mathcal{V}^* = \text{SIGN}(U)$, for a linear subspace $U \subseteq \mathbb{R}^n$.

For the following, choose $\mathcal{V}^*$ to arise from any one of the above models: perhaps it is best to take $\mathcal{A} = \{H_1, \dots, H_n\}$ to be an oriented, essential arrangement of hyperplanes, and let $\mathcal{V}^*$ be the set of sign vectors of all its cones. (For example, take the one sketched in the proof of Theorem 7.16.)

We need the following operations on sign vectors. The *zero vector* $\mathbf{0}$ and the *negative* $-\mathbf{u}$ of a sign vector $\mathbf{u}$ have their obvious meanings. The *support*